

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		maximum possible mass = 47.2 (3) if incorrect credit following steps in reacting masses calculation 117 of NaCl gives 46 of Na (1) 120 of NaCl gives $\frac{46}{117} \times 120$ of Na (1) ecf possible accept alternative method using moles $n(\text{NaCl}) = 2051$ (1) $n(\text{Na}) = 2051$ (1) $m = 2051 \times 23 = 47173$ $m = 47.2$ (1) percentage yield = 80.6 / 81 / 80.65 (1) ecf possible						
		(ii)		accept any of following for (1) - sodium chloride used was impure - not all the sodium chloride had reacted - side reactions taking place - loss of product	1			1		1
		(iii)		sodium would react with any water present		1		1		1